

Deepfake Detection using Multi-Stream Convolutional Neural Networks with Attention-based Fusion

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Abstract—In recent years, DeepFake technology has rapidly evolved by enabling the development of highly realistic forged facial videos that pose serious threats to digital trust and security. Traditional deep learning approaches, although effective in image classification, usually struggle to generalize against subtle manipulation artifacts, compression noise, and temporal inconsistencies. To address these challenges, a Multi-Stream Convolutional Neural Network (MS-CNN) framework is proposed for robust deepfake detection. Initially, utilizes the FaceForensics++ dataset which is a widely recognized benchmark containing authentic and manipulated facial videos. Preprocessing incorporates face detection, alignment, and the generation of complementary modalities, which include frequency-domain representations, residual noise maps, and temporal signals, such as rPPG and optical flow. These multi-modal inputs were passed through parallel CNN streams for feature extraction to capture spatial, spectral, and temporal inconsistencies. The extracted features are combined using an attention-based adaptive fusion mechanism that dynamically learns the relative importance of each modality. Finally, a fully connected softmax classifier produces end-to-end predictions that differentiate between real and fake videos. The experimental results show that the MS-CNN obtains higher accuracy (99.77%), precision (98.95%), recall (98.45%), and F1-score (98.76%) when compared with the existing CNN model, which improves detection robustness and generalization in different manipulation techniques.

Keywords—deepfake detection, multi-stream convolutional neural networks, residual noise maps, softmax classifier, temporal signals

I. INTRODUCTION

Recently, the creation of highly convincing realistic fake digital media has been increasing rapidly using advanced deep generative models such as Stable Diffusion, Sora [1]. These fake synthetic media are referred to as deepfakes, which include images, videos, audio, and text that are nearly unidentifiable from real content. As deepfake generation tools are easily available, anybody can create fake media without any advanced technical proficiency, which increases cyber-crime by spreading misinformation and threats daily [2]. Therefore, the development of a robust technology for

deepfake detection is crucial. Although various technologies have been developed, the detection of deep fake media remains a difficult task [3]. Many approaches have been introduced by generative models for detecting artifacts or statistical misdeeds; however, signs can be difficult to spot as generative models advance, creating a nonstop cat-and-mouse battle between deepfake creators and detectors. The generalization of data is challenging for traditional detection methods, as there are some techniques that cause huge chances of overfitting to training datasets and perform poorly against new-generation methods [4]. Furthermore, most traditional models are computationally expensive, which makes the model challenging for real-world applications on mobile or resource-constrained devices [5]. All of these limitations lack trust and practical usability, mostly in forensic or security applications. As deepfake technologies are rapidly increasing the misuse of sensitive information in large-scale data, these challenges have motivated the development of efficient detection frameworks for digital media environments [6]. In this context, contrastive learning-based general deepfake detection with multi-scale RGB frequency clues has shown promise in improving generalization and robustness [7]. The novel system not only identifies between fake and real, but also delivers insights into which modality has been manipulated to increase trust and usability in real-world applications [8]. Additionally, with the growing use of deepfakes in video-based misinformation campaigns, the effective fusion of audio and visual modalities is essential for capturing complementary signals [9]. The main goal should be to decrease the gap between existing unimodal and multi-modal detection approaches by providing effective solutions that also work against unseen deepfake methods, operate efficiently on real-world platforms, and enhance decision-making abilities [10]. Deepfake detection methods primarily use Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to capture pixel-level artifacts, compression noise, and global inconsistencies. By integrating local and global features, some machine learning models such as Xception and hybrid CNN-ViT increase performance, whereas auxiliary modalities such as frequency signals, residual noise maps, and biological cues trace subtle manipulation. Recently, multi-stream and audio-visual fusion

approaches have been introduced to influence complementary information; however, they face issues in generalization to new operations and fusing multi-modal features. These challenges emphasize the need for more adaptive and efficient fusion technologies for robust deep fake detection. The key contributions of this study are as follows:

- A multi-stage preprocessing strategy is planned, including face detection and alignment, frequency representation using Discrete Cosine Transform (DCT), cleaning for residual noise via Spatial Rich Model (SRM) filters, and constructing temporally varying features via optical flow and rPPG. This transactional pipeline allows for the preservation of fine spatial, spectral, and physiological artifacts produced during deepfake generation, and reduces noise.
- A Multi-Stream Convolutional Neural Network (MS-CNN) was selected to allow processing of RGB, frequency, residual, and temporal workflows in parallel mode, which is an attention-based adaptive fusion approach that allows the multi-stream framework to adaptively weigh the features, improving robustness directly related to the various effects of manipulation.
- The Softmax classifier at the end of the network enables more suitable end-to-end training. The design facilitates better accuracy, allows for more interpretability with a probabilistic output, and ensures computational efficiency to allow for suitable deployment in the world, including TinyML integration.

The remainder of this paper is organized as follows: Related works and their drawbacks are given in Section 2; the overall explanation of the proposed methodology is described in Section 3; the experimental results and discussion are presented in Section 4; and Section 5 concludes the paper.

II. LITERATURE REVIEW

Chunmei Duan and Taochuan Zhang. [11] developed Surface defect classification and detection using a Multi-Stream Deep Convolutional Neural Network (CNN). The data collection consisted of Polyethylene Terephthalate (PET), which has defects, and defect groups are scratches, bubbles, and spots. In feature extraction, convolutional layers are used to extract the edge, color, and texture features, and a max-pooling layer is used to extract the relevant features. Feature fusion was used to extract different features and classify the final results. Google Net and VGG were used to train the deep CNN and extract features from the original image. Google Net operates the convolutional layer and inception block, and VGG generates the gradient image to detect the object. The advantages of this model are its good stability, accuracy, and convergence. However, on-site data on production and collected sample images of high quality may be worse.

Zhang et al. [12] developed a deep multistream neural network for automatic modulation classification. The data collection consists of a signal dataset that has schemes of modulation, zero mean, bandlimited, and Gaussian white noise. In data preprocessing, the input signals are divided into different lengths to improve classification. For feature extraction, a convolutional layer is used to extract signal characteristics, and global average pooling is used to extract

relevant features. A Multi-Stream Wide Network (MSWN) extracts signal features through a convolutional layer and multiple dimensions of reaggregation for strong correlation. Superposition Convolutional Unit (SCU) reduces the features using one convolutional layer and uses nonlinear features for signal data to improve the classification effect. The advantage of this model is that it has a low parameter size and is useful for real-time scenery. However, the model requires huge computational complexity for training.

R. Ragavendra and Brindha Subburaj. [13] developed fuzzy system-based decision fusion and spatiotemporal-structural anomaly learning for deep fake detection. The data collection consisted of face forensics and celeb-df, which had fake, face2face, and manipulated videos. During data preprocessing, data stored in the directory and frames are in the Joint Photographic Expert Group (JPG) format. For feature extraction, if a face is detected in a frame, it is cropped in the facial region using the You Only Look Once (YOLO) model. In the methodology, Fuzzy Deepfake Detection (FDD) uses three ResNet-50 classifiers and is trained on edge, optical flow, and RGB to learn different characteristics of the frame, and a fusion technique is used to combine the results of the three classification models. The advantage of this model is that it has diverse and temporal characteristics for better identification results. However, the model had some inconsistencies in the learning of the sample images.

Jiann-Shiun Yuan and Aminollah Khormali. [14] developed a vision transformer for end-to-end deep false detection. The data collection consisted of WildDeepfake, Celeb-DF, and face forensics, which have deepfake, face2face, high-quality deep fake videos, and both original and deep fake videos, respectively. In data preprocessing, deepfake techniques are used in the facial areas of scattered regions and trained mainly on the face area to reduce complexity. For feature extraction, the local structures and patches were extracted using patch embeddings. In the methodology, the transformer mainly uses four techniques: embedding, path extraction, transformer block, and attention-based multi-classifier to detect the deepfake classification purpose. The transformer uses both the local and global features of images and differs from forgery scales. The advantage of this model is that it can effectively capture various deep fakes. However, the model techniques are limited to facial and non-background areas.

Harinadh et al. [15] developed a deep learning detection image forery using a Fusion-Based Approach. The data collection consists of FaceForensics++, IMD2020 and CASIA, which has deep fake, copy move and manipulated images. In data preprocessing for the consistency of datasets, the compression and resolution differences were reduced. In feature extraction, the texture and spatial-based features in the images are extracted to reduce the model complexity. In methodology, multi-modal fusion detects texture and color inconsistencies using a Convolutional Neural Network (CNN) and examines the mismatches of images. CNN trained on different types, including copy, deepfake, forgeries, and splicing, are also used for classification purposes. Finally, the CNN predicts whether an image is authentic or forged. The advantage of this model is that it is suitable for real-world application. However, this model has high false-negative and false-positive rates.

III. METHODOLOGY

The proposed approach uses a Multi-Stream Convolutional Neural Network (MS-CNN) for deepfake detection, where spatial, spectral, residual, and temporal cues are integrated. Preprocessing steps include face detection and alignment, frequency analysis through Discrete Cosine Transform (DCT), creating residual noise maps using Spatial Rich Model (SRM) filtering, and temporal signals created using optical flow from the temporal component and remote

photoplethysmography (rPPG). Each of these modalities has its own dedicated CNN stream, where features are extracted from each stream. An attention-based fusion mechanism fuses all the extracted features, where it places more weight on the most discriminative cues. Finally, a softmax classifier calculates the probability of the video being real versus fake to produce an accurate and robust detection for manipulated videos. The flow diagram of the proposed MS-CNN is shown in Figure 1.

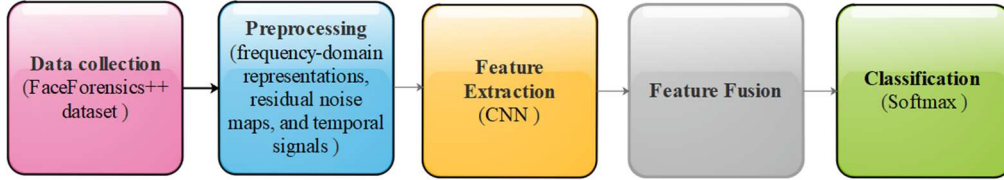


Fig. 1. Flow diagram of the proposed MS-CNN framework.

A. Dataset

In this research, the FaceForensics++ (FF++) dataset [11], which is the most commonly used benchmark for deepfake detection researchers. FF++ comprises thousands of real and altered facial videos manipulated by DeepFakes, FaceSwap, Face2Face, and NeuralTextures, produced with several levels of compression for robustness and variability in evaluation. The higher-quality subset of FF++ (there exist higher-quality and lower-quality subsets) normalizes the data size between trials by extracting video frames consistently over time (i.e., extracting 10 frames from each video to ensure a robust subject-independent testing evaluation and minimizing variability in training validation and test subjects). The extracted frames were processed for face detection, alignment, and normalization to a size of 224×224 pixels. The dataset was separated into subject-independent training (70%), validation (15%), and testing (15%) evaluations.

B. Preprocessing

Preprocessing is extremely important in preparing the FaceForensics++ for a deepfake detector. Forgeries often come with subtle anomalies in the spatial, frequency, and temporal domains, and in ditching the perturbations that reconstruct it, careful preprocessing allows the MS-CNN to receive conventional and meaningful inputs. preprocessing has five components: face detection and alignment, frequency transformation, extraction of the residual noise, construction of a temporal signal, and data augmentation.

1) *Face Detection and Alignment*: Videos are sampled into frames and MTCNN, and a series of cascaded convolutional layers are used to identify facial regions, providing a robust detection solution in terms of pose and lighting. The MTCNN provides bounding box detection and then utilizes facial landmarks from the eyes and mouth region to refine the bounding boxes. An affine transformation is then utilized to geo-align the faces, which reduces the effect of the head pose or rotation. In addition, the cropped faces are then resized from the video to uniform 224×224 pixels to ensure consistent input between the streams.

2) *Residual Noise Extraction (SRM Filtering)*: Frequency-domain analysis provides an essential means of identifying global inconsistencies in manipulated photographs, and residual filtering detects pixel-level incongruences. These types of pixel-level artifacts are

derived from the rapid generation or compression of deepfake images and are mainly stable artifacts that are likely invisible in RGB and frequency maps. To visualize these incongruences, we adopted a spatial-rich model (SRM) high-pass filter. Spatial Rich Model (SRM) filters are commonly used in digital image forensics, as shown in Equation (1):

$$K = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{bmatrix} \quad (1)$$

This kernel emphasizes high-frequency details by suppressing the smooth image regions while amplifying local variations. The residual map is generated by convolution, as shown in Equation (2).

$$R(x, y) = (I * K)(x, y) \quad (2)$$

Where I denotes the input facial image. The residual enhances edges, inconsistent textures, interpolation noise, and boundary inconsistencies, all of which signal potential forgeries. Residual noise maps are better at finding previously unseen artifacts at the microscopic level, such as blending, resampling, and other smoothing inconsistencies, whereas frequency-based analyses look for irregular compressions. Even the noise tracked in the noise map may become subperceptual; however, they are strong forensic markers, regardless of whether they clearly separate the real from synthetic. By providing the residual map as a separate input stream, MS-CNNs can learn to identify localized manipulation outcomes more efficiently.

3) *Temporal Signal Construction*: Deepfake manipulations often compromise the temporal consistency, producing unnatural motion transitions or missing physiological signals. To address this limitation, we extracted two temporal features: optical flow and remote photoplethysmography (rPPG), which serve as complementary forensic indications. Optical Flow estimates the displacement of pixel intensities between consecutive frames, capturing abnormal facial dynamics, such as jitter, flickering, or misaligned lip movements. It is formulated using the brightness constancy constraint, as shown in Equation (3).

$$I_x u + I_y v + I_t = 0 \quad (3)$$

Where I_x and I_y are the spatial intensity gradients, I_t is the temporal gradient, and (u, v) is the horizontal and vertical flow vectors. In genuine videos, flows would develop smoothly and coherently, while fake videos are likely to express flows of irregular motion or discontinuous motion fields. Remote photoplethysmography (rPPG) relies on small variations in color in one of the skin regions, usually extending across the green channel. In real videos, these appear as periods. In fake videos, it is hypothesized that they are entirely distorted or eliminated. Processed rPPG signals were separated into 2D temporal feature maps by chunking time-series traces into fixed windows of seconds and spatial stacking. By merging the optical flow and rPPG maps, the preprocessing phase conserves motion coherence, and physiological (i.e., "real-life") integrity at the same time, delivering strong temporal features for the MS-CNN.

4) *Data Augmentation*: To further increase the resilience and generalizability of the specified MS-CNN, different forms of data augmentation were used during the preprocessing phase. Augmentation allows the model to adapt to a range of real-world conditions (including its loss due to compression, sensor noise, and fluctuations due to the environment). Attaining augmentation begins with the introduction of randomly generated JPEG compression with varying quality factors to replicate the commonly observed conditions on social media platforms for lossy storage and transmission. Second, Gaussian noise is injected to replicate the errors associated with sensor noise in low-quality recordings at this level. Third, Gaussian and motion blurring are applied to replicate common issues with camera shaking, and defocusing is often present in uncontrolled conditions (i.e., everyday recording). Fourth, brightness and contrast were varied to replicate common variations in lighting conditions, including varying shadows and disparities in exposure under different recording conditions. Finally, horizontal flipping was used as a geometric transformation to replicate the variation in facial orientations, thereby avoiding overfitting to particular poses. Formal augmentation can be described as shown in Equation (4).

$$I' = \mathcal{A}(I) \quad (4)$$

Where I is the original image and $\mathcal{A}$ is a stochastic augmentation operator drawn from the transformation set. By applying stochastic augmentations, the dataset becomes more representative of its real conditions, which guarantees that once trained, the model will have significantly better robustness and generalization to manipulations that it has not seen before.

C. Feature Extraction (MS-CNN)

Once preprocessing is completed, the input modalities, RGB images, frequency-domain maps, and residual noise maps, as well as the temporal features, are then fed into the MS-CNN for feature extraction. Each modality was assigned to its own specific CNN stream, allowing the network to learn, encode, and extract distinct forensic cues. RGB Stream further extracts semantic and spatial patterns, such as face structures and discrepancies in expressions. The frequency stream learns using DCT-transformed inputs to determine compression

artifacts and blending errors. The residual stream focused on pixel-level discrepancies and unnatural texturing. Finally, the temporal stream learns the motion dynamics from the optical flow and physiological coherence from the rPPG-derived maps. From each stream, the model goes through multiple layers during feature extraction by converting into 1-minute somatic information, which can be mathematically defined in Equation (5):

$$F_{i,j}^k = \sigma(\sum_m (W_m^k * X_m)_{i,j} + b^k) \quad (5)$$

Where X_m is the input, W_m^k is the kernel, b^k is the bias, and σ is the activation function (ReLU). This process produces hierarchical feature maps that progressively obtain low-level edges for high-level semantics. Batch normalization and pooling layers were included to stabilize the training and reduce peak dimensionality. In contrast to single-stream CNNs, MS-CNN allow different streams to learn complementary features across multiple domains simultaneously. This multiscope perspective produces spatial, spectral, residual, and temporal anomalies that are simultaneously represented. During feature fusion, the outputs of the individual streams were subsequently aligned to share a latent space and represent the full embedding of the input sample. Consequently, MS-CNN provides a comprehensive and discriminative representation of the presented data, serving as a mechanism for deep fake detection, as discussed in this work.

D. Feature Fusion

After obtaining feature embeddings independently from the MS-CNN RGB, the frequency, residual, and temporal streams of features are integrated to form a robust representation for classification. Instead of some static measures, as done in the proposed attention-based adaptive fusion, it is essential to dynamically generate weights for the importance of the embedding (and features) between each modality and allow the system to focus on the most difficult and discriminative features given an input. Let the feature embeddings from each stream be labeled as. The embeddings are first projected into a common latent space and then concatenated into a joint representation, as shown in Equation (6):

$$F = [f_{rgb} \| f_{freq} \| f_{res} \| f_{temp}] \quad (6)$$

An attention mechanism then computes the normalized weights for each modality, as shown in Equation (7):

$$\alpha_i = \frac{\exp(W_a f_i)}{\sum_j \exp(W_a f_j)} \quad (7)$$

Where W_a are learnable parameters and α_i represents the attention score of modality i . The final fused feature is expressed as shown in Equation (8):

$$F_{fused} = \sum_i \alpha_i f_i \quad (8)$$

This adaptive fusion approach guarantees that the network favors residual features when there are inconsistencies in noise when there are motion artifacts. Stream dropout is also used when training and randomly masks one modality; therefore, the network does not over-rely on a single stream and is more robust. Therefore, the fused representation

features complementary spatial, spectral, and temporal cues, forming a discriminative embedding that passes into the final classification stage.

E. Classification

The final stage of the proposed framework involves the classification of fused features into their respective categories: real or false. After the feature fusion, the aggregated representation F_{fused} contains rich and complementary information from the RGB, frequency, residual, and temporal domains. However, this high-dimensional vector must be mapped into a decision space to enable robust discrimination. To achieve this, we employed a fully connected neural network followed by a softmax classifier, which offers an efficient and differentiable end-to-end classification mechanism. Initially, the fused feature vector F_{fused} is passed through a dense layer to reduce dimensionality while preserving discriminative power, as shown in Equation (9):

$$h = \sigma(W_f F_{\text{fused}} + b_f) \quad (9)$$

Where W_f and b_f denote learnable weights and biases, respectively, and σ represents the activation function, typically a ReLU. Dropout regularization was applied to prevent overfitting, whereas batch normalization ensured stable gradient propagation. This transformation enables the network to focus on the most informative feature components and discard redundant components. The transformed representation is then passed to the softmax layer, which converts the logits into probability distributions over target classes. The Softmax function is mathematically expressed as Equation (10):

$$P(y = i | x) = \frac{\exp(z_i)}{\sum_{j=1}^C \exp(z_j)} \quad (10)$$

Where z_i is the logit corresponding to class i , and C is the number of classes. Because deepfake detection is a binary classification problem, $C = 2$, corresponding to the labels Real and Fake. The final predicted class is obtained using Equation (11):

$$\hat{y} = \arg \max_i P(y = i | x) \quad (11)$$

This probabilistic formulation ensures that the model not only outputs a class label, but also provides confidence values for interpretability. For instance, a high probability for the "Fake" class signals strong evidence of manipulation. Because Softmax is fully differentiable, gradients are backpropagated through both the classifier and earlier CNN streams, thereby enabling end-to-end training. This allows the network to fine-tune the feature extraction jointly with classification, thereby improving the overall detection accuracy. Moreover, the lightweight design of the fully connected layers ensures computational efficiency, making the framework adaptable to TinyML deployment in mobile or IoT-enabled systems. Thus, the Softmax classifier not only strengthens classification robustness but also ensures scalability and real-world applicability.

IV. EXPERIMENTAL RESULTS

In this study, an efficient multistream convolutional neural network (MS-CNN) framework is presented with robust

deepfake detection by implementing the Python package TensorFlow. For this purpose, software and hardware supplies were used for experimental analysis. The software requirements include operating system windows 11 pro and hardware requirements such as Intel Core i5 processor with 16GB RAM and 3.63 GHz processing speed. For the evaluation of the proposed MS-CNN, the accuracy, precision, recall and F1-score were utilized as the evaluation metrics, which are expressed in Equations (11–14):

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (11)$$

$$\text{Precision} = \frac{TN}{TP+FP} \quad (12)$$

$$\text{Recall} = \frac{TP}{TP+FN} \times 100 \quad (13)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (14)$$

TP, True Positive, TN, True Negative, FP, False Positive and FN, False Negative.

A. Performance analysis

The performance analysis of the proposed MS-CNN for multi-stage wheat yield prediction was performed using various traditional models, namely, recurrent neural networks (RNN) and gated recurrent units (GRU), with the help of evaluation metrics. The performance evaluation of the proposed MS-CNN is described in Figure 1.

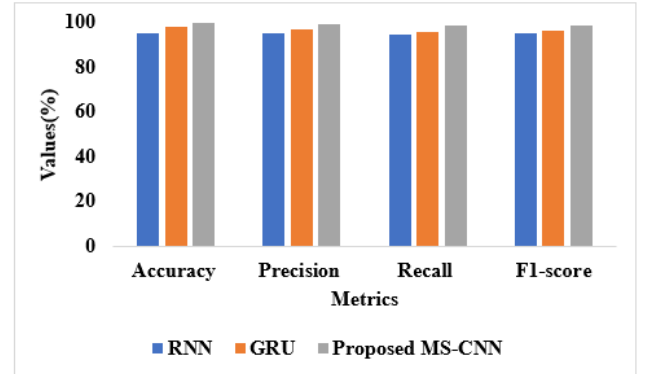


Fig. 2. Performance analysis of proposed MS-CNN.

Figure 1 presents a performance analysis of the proposed MS-CNN with various traditional models, including RNN and GRU. The proposed MS-CNN provided better results in terms of accuracy, precision, recall and F1-score are 99.77%, 98.95%, 98.45%, 98.76% respectively.

B. Comparative analysis

A comparative evaluation of the proposed MS-CNN was performed by comparing it with an existing Deep CNN [11] model. The proposed MS-CNN is compared with the evaluation metrics of accuracy, precision, recall, and F1-score, as explained in Table 1.

TABLE I. COMPARATIVE ANALYSIS OF THE PROPOSED MS-CNN.

Comparative models	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Deep CNN[11]	98.3	97.8	97.6	97.7
Proposed MS-CNN	99.77	98.95	98.45	98.76

Table 1 shows a comparative evaluation of the proposed MS-CNN with the existing Deep CNN [11] by the evaluation metrics. MSD-CNN provided the best results in terms of accuracy (98.3%), precision (97.8%), recall (97.6%), and F1-score (97.7%). The results show that the proposed MS-CNN outperformed the existing models by using an attention-based adaptive fusion mechanism for learning the relative importance of each modality. The proposed MS-CNN achieved better results in terms of accuracy (99.77%), precision (98.95%), recall (98.45%), and F1-score (98.76%).

C. Discussion

The main objective of this paper is to introduce a model that detects deepfakes using Multi-Stream Convolutional Neural Networks with Attention-Based Fusion. The existing RNN and GRU, and Deep CNN [11] models are effective for image classification; however, they often struggle to generalize against subtle manipulations, compression noise, and temporal inconsistencies in deepfake detection. The RNN model works on sequential data, but is highly vulnerable to vanishing gradients and long-term dependency learning. To solve this issue, the GRU model introduces gating mechanisms. However, it struggles to handle highly complex patterns. Furthermore, the Deep CNN model solves this problem by capturing hierarchical spatial features; however, it generates higher computational cost and scalability issues. Hence, a Multi-Stream Convolutional Neural Network (MS-CNN) model is introduced, which integrates multiple feature streams to efficiently capture spatial-temporal representations. This method decreases the computational overhead, enhances scalability, and overcomes the limitations caused by RNN, GRU, and MS Deep CNN models. Overall, the MS CNN model provides a balanced solution that strengthens security, while making it efficient and suitable for modern digital environments.

V. CONCLUSION

The proposed MS-CNN framework is a powerful, intelligent, and scalable system for robust deep fake detection in smart digital environments. The combination of multistream convolutional neural networks (MS-CNN) with attention-based fusion offers a system in which robust feature extraction, temporal consistency, manipulation detection, and classification accuracy are handled for deepfake detection. The MS-CNN architecture layers capture diverse features from multiple streams for attention-guided fusion and final classification, which can solve the critical challenges in deep fake analysis. Moreover, this model works efficiently on subtle facial manipulations, compression artifacts, and temporal inconsistencies, whereas its attention mechanism ensures that the most discriminative cues are emphasized. The experimental results indicate an improved detection accuracy with reduced computational cost, which makes the proposed methodology effective and scalable for real-world applications. The proposed MS-CNN achieved better results in terms of accuracy (99.77%), precision (98.95%), recall (98.45%), and F1-score (98.76%). In future work, the MS-CNN framework will be advanced by combining transformer-based temporal modeling for a better temporal understanding and adversarial training to reinforce the detection procedure of deepfake techniques.

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